TANGENTS, COTANGENTS, DERIVATIONS, AND DIFFERENTIALS

1. Introduction

In differential geometry, one of the first constructions on a smooth manifold is that of the tangent space T_pM at a point p, and soon after that we construct the tangent bundle TM. Tangent spaces are very useful when looking at maps between manifolds, because the differential of a map is essentially a linearization of the map, and the rank of the differential tells us a lot of useful information. Unfortunately, in algebra we do not have the tools available in the smooth category (such as partitions of unity), so we have to try to define tangents in an entirely algebraic way.

Unlike differential geometry, where singularities are rare and difficult to deal with, singularities show up in algebraic geometry all the time, but the rigidity of our algebraic tools means that we can analyze a variety at its singular points. One of the main benefits of generalizing the definition of tangents is that we can use them in situations that are singular.

For example, consider the *cuspidal cubic*, the curve cut out by the equation $y^2 = x^3$ in $\mathbb{A}^2_k$. If we sketch this out, it has a cusp at the origin, where it is non-differentiable. If we consider lines tangent to the curve at any point that is not the origin, there is just one line, but if we look at the origin it there is a 2-dimensional space of lines that can be thought of as being tangent to the curve. In the next problem we make some computations that will motivate the equivalences we will prove in this worksheet.

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(1) **Motivating example**. Let $k = \mathbb{C}$, and let $A = k[x,y]/\langle x^2 - y^3\rangle$. Then $\mathrm{Spec}(A)$ is the *cuspidal cubic*, and we can visualize it as (the closed points of) $\mathbb{V}(\langle x^2 - y^3\rangle) \subseteq \mathbb{A}^2_k$.

   (a) For any maximal ideal $\mathfrak{m} \neq \langle x, y\rangle$, show that $\mathfrak{m}/\mathfrak{m}^2 \cong k$ but $\langle x,y\rangle/\langle x,y\rangle^2 \cong k^2$

   (b) Let $f = x^2 - y^3$, with formal partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. For a point p lying on the curve in k^2, what is the rank of the Jacobian matrix $J(p) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}$?

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2. Derivations and Tangents

In differential geometry, one way of defining tangents at a point p of a smooth manifold M is by looking at point derivations of $C^\infty(M)$ at p. These are linear maps $D : C^\infty(M) \to \mathbb{R}$ that satisfy $D(fg) = f(p)D(g) + g(p)D(f)$. An example of this, given $M = \mathbb{R}^n$, is the directional derivative D_v for $v = \sum_i v_i x_i$. The tangent vector can be identified with the derivation $D_v f = \sum_i v_i \frac{\partial f}{\partial x_i}$. We adopt a similar approach to defining tangents on the spectrum of a ring.

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To extend this analogy with the ring of smooth functions on a smooth manifold, we treat a ring A as a ring of functions on its spectrum $\mathrm{Spec}(A)$. We will call an element $f \in A$ a *function* on $\mathrm{Spec}(A)$, and to evaluate it at a point $p = [\mathfrak{p}] \in \mathrm{Spec}(A)$, we reduce it modulo $\mathfrak{p}$ and consider it an element of the residue field $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong \mathrm{Frac}(R/\mathfrak{p})$. In the case that $\mathfrak{m}$ is a maximal ideal, $f([\mathfrak{m}]) = f \pmod{\mathfrak{m}}$. This notion of evaluating a function strictly extends the case of reduced k-algebras: If $\mathfrak{m}$ is a maximal ideal in a k-algebra for an algebraically closed field k, then $\mathfrak{m}$ corresponds to a point of k^n and $f \pmod{\mathfrak{m}}$ is the value of f at that point. If k is *not* algebraically closed, then $\mathfrak{m}$ can correspond to (a Galois orbit of) a point in some extension of k, but in that extension reducing f modulo $\mathfrak{m}$ is still evaluation. Note, however, that if our ring is not reduced, functions are *not* determined by their values at all points. Since we will mainly work with integral k-algebras, we will not have to worry about this.

Let A be a k-algebra, and $\mathfrak{m}$ a maximal ideal. A *derivation* of A at $\mathfrak{m}$ is given by a k-linear map $D : A \to \kappa(\mathfrak{m})$ satisfying $D(fg) = f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f)$. The set of such derivations is denoted $\mathrm{Der}_k(A, \kappa(\mathfrak{m}))$ and is vector space over k. Later, it will be useful to consider a more general definition of a derivation: given a ring R, an R-algebra S, and an S-module M, a *derivation* over R from S to M is an R-linear map D satisfying $D(fg) = f \cdot D(g) + g \cdot D(f)$; here, f and g are acting on elements in M via the module structure given to M. Our earlier definition is a special case of this definition, where the module structure on $\kappa(\mathfrak{m})$ is given by $f \cdot m = f(\mathfrak{m})m = \pi(f)m$, where π is the natural projection $A \twoheadrightarrow A/\mathfrak{m}$. The set of such derivations is denoted $\mathrm{Der}_R(S, M)$ and is a module over S.

There are other ways to define tangent vectors, and we will see they are all equivalent. Given a maximal ideal $\mathfrak{m}$, $\mathfrak{m}/\mathfrak{m}^2$ is a vector space over the residue field $\kappa(\mathfrak{m})$, and it turns out that this is dual to the vector space of derivations at $\mathfrak{m}$. Later, we will see that for varieties over algebraically closed fields, the Jacobian criterion can be used to identify derivations with vectors in the kernel of the (algebraically defined) Jacobian matrix, exactly agreeing with our intuition from smooth manifolds. Lastly, we will look at a scheme-theoretic way of looking at a tangent vector: the scheme $\mathrm{Spec}(k[\epsilon]/\langle \epsilon^2 \rangle)$ has just one point, $\langle \epsilon \rangle$, but it also has a "one-dimensional fuzz" because of the nilpotent ϵ. A map from this scheme to $\mathrm{Spec}\, A$, defined as a map from $A \to k[\epsilon]/\langle \epsilon^2 \rangle$, singles out exactly the data of a closed point of $\mathrm{Spec}(A)$ together with a derivation at that point.

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(2) **The Zariski Cotangent space**. Let R be a ring. For any closed point p of $\mathrm{Spec}(R)$, let $\kappa(p)$ denote the residue field.

(a) Let I be any ideal. Recall (or show) that for any R-module M, $M/IM \cong M \otimes_R R/I$, and thus $I/I^2 \cong I \otimes_R R/I$. Show that this gives I/I^2 the structure of an R/I module.

(b) Suppose $p = [\mathfrak{m}]$ is a closed point of $\mathrm{Spec}(R)$. Show that $\mathfrak{m}/\mathfrak{m}^2$ is a vector space over $\kappa(p)$.

(c) Let $R_{\mathfrak{m}}$ be the localization at $\mathfrak{m}$. Show that $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{m}R_{\mathfrak{m}}/(\mathfrak{m}R_{\mathfrak{m}})^2$

(d) Show that if $\mathfrak{m}$ is finitely generated, then $\mathfrak{m}/\mathfrak{m}^2$ is finite dimensional (Hint: Nakayama's lemma).

(3) **Point derivations**. Let A be a k-algebra and $\mathfrak{m}$ a maximal ideal of A. Let D be a derivation at $\mathfrak{m}$, i.e. a k-linear map $A \to k$ such that $D(fg) = f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f)$.

(a) Show that if $f \in k$ i.e. f is a "constant function," then $D(f) = 0$

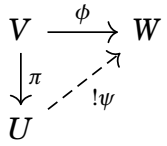
(b) Show that if f is idempotent, i.e. $f^2 = f$, then $D(f) = 0$

(c) For a ring R, an R-algebra S, and an S-module M, we can define $\mathrm{Der}_R(S, M)$ to be the set of R-linear maps $D : S \to M$ such that $D(fg) = fD(g) + gD(f)$. Setting $R = k, S = A, M = k$, we can recover our point derivations. If $\kappa(\mathfrak{m}) = \kappa(\mathfrak{n}) = k$, is $\mathrm{Der}_k(A, k)$ at $\mathfrak{m}$ isomorphic to $\mathrm{Der}_k(A, k)$ at $\mathfrak{n}$?

(4) **Point derivations are dual to Zariski cotangents.** Let A be a k-algebra and $[\mathfrak{m}]$ a k-rational closed point i.e. $A/\mathfrak{m} = k$.

(a) Show that $A \cong k \oplus \mathfrak{m}$ as k-vector spaces.

(b) Recall (or show) the factoring-through principle: If $\phi : V \to W$ is a linear map of k-vector spaces and $\pi : V \to U$ is a surjective linear map such that $\ker \pi \subseteq \ker \phi$, then ϕ factors uniquely through π, i.e. there exists a unique map $\psi : U \to W$ such that $\phi = \psi \circ \pi$:



(c) Show that given a derivation $D : A \to k$ at $\mathfrak{m}$, there is a unique way of associating to it an element of $(\mathfrak{m}/\mathfrak{m}^2)^\vee$, and that this association is linear

(d) Complete the proof that $\mathrm{Der}_k(A, k) \cong (\mathfrak{m}/\mathfrak{m}^2)^\vee$ by showing the surjectivity of the above assignment

(5) **Tangents and the dual numbers**. Let k be a field and R a k-algebra. Let $\pi, \psi : k[\epsilon]/\langle\epsilon\rangle^2 \to k$ be the k-linear maps given by $\pi(a + b\epsilon) = a, \psi(a + b\epsilon) = b$. Let $\phi : R \to k[\epsilon]/\langle\epsilon\rangle^2$ be a ring homomorphism and $\phi^\# : \mathrm{Spec}(k[\epsilon]/\langle\epsilon\rangle^2) \to \mathrm{Spec}\, R$ be the induced map on spectra.

(a) Show that $\phi^\#(\langle\epsilon\rangle)$, the image of the unique point in $\mathrm{Spec}(k[\epsilon]/\langle\epsilon^2\rangle)$, is a maximal ideal $\mathfrak{m}$ with residue field k

(b) Show that $\psi \circ \phi$ is a k-linear derivation of R at $p = [\mathfrak{m}]$

(c) Show that given a k-rational point p and a derivation D of R at p, there is a ring homomorphism from R to $k[\epsilon]/\langle\epsilon\rangle^2$

(d) Conclude that maps from $\mathrm{Spec}(k[\epsilon]/\langle\epsilon^2\rangle) \to \mathrm{Spec}(R)$ (induced by a ring map) are precisely the data of a k-rational point of $\mathrm{Spec}(R)$ along with a tangent vector at that point.

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3. Kähler differentials and Cotangents

We have now found a way to describe tangent vectors algebraically, but other than the fact that these definitions agree for smooth manifolds, we don't have much reason to believe that our definition works as intended. In this section, we introduce a tool to analyze derivations, called the *module of differentials*. This plays the role in algebraic geometry that the cotangent bundle plays in differential geometry. Our goal in this section is to prove the *Jacobian criterion*: the smoothness of an algebraic variety can be computed using the Jacobian matrix, exactly like in differential geometry!

We return to the general case of derivations $\mathrm{Der}_R(S,M)$ from an R-algebra S to an S-module M. We want to linearize derivations, so we construct a "universal derivation." To do so, let $\Omega_{S/R}$ consist of the free S module with generators the formal symbols $d(f)$ (which we will write as df if there is no ambiguity) for $f \in R$, subject to the linearity relations $d(af+bg) = a\,df + b\,dg$ for $a,b \in R, f,g \in S$, and the Leibniz relations $d(fg) = g\,df + f\,dg$ for $f,g \in S$. This $\Omega_{S/R}$ is called the *module of differentials*, or the module of Kähler differentials. By construction, $d : S \to \Omega_{S/R}$ is an R-derivation, and moreover, it is the universal R-derivation: given an R-derivation $D : S \to M$, there is a unique map $\overline{D} : \Omega_{S/R} \to M$ such that the diagram

$$\begin{array}{ccc} S & \xrightarrow{D} & M \\ \downarrow{\scriptstyle d} & \nearrow_{!\overline{D}} & \\ \Omega_{S/R} & & \end{array}$$

commutes.

The module of differentials behaves very nicely. For polynomial rings over fields i.e. $R = k, S = k[x_1,\ldots,x_n]$, it is given by adjoining dx_i to S i.e. $\Omega_{S/R} = \bigoplus_i S\,dx_i$. It commutes with localization via a quotient rule, and it can even be turned into a nice functor on pairs of rings (though we will not go that far).

An important tool for computations involving Kähler differentials is the conormal sequence: given a surjective map of R-algebras $\pi : S \to T$ with kernel I, there exists an exact sequence of T-modules

$$I/I^2 \xrightarrow{d} T \otimes_S \Omega_{S/R} \xrightarrow{D\pi} \Omega_{T/R} \longrightarrow 0$$

with the left map sending $[f] \mapsto 1 \otimes df$ and $D\pi$ sending $c \otimes db \mapsto c\,db$

An important special case of this is given by $\pi : A \to k$ when A is a k-algebra, $\mathfrak{m}$ a k-rational point, and π the projection map $A \to A/\mathfrak{m}$. Then, $R = k, S = A, T = k$ and $I = \mathfrak{m}$. Since $T \otimes_A \Omega_{S/R} = A/\mathfrak{m} \otimes \Omega_{A/k} = \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$, and $\Omega_{T/R} = \Omega_{k/k} = 0$, the sequence turns into

$$\mathfrak{m}/\mathfrak{m}^2 \xrightarrow{d} \Omega_{A/k}/\mathfrak{m}\Omega_{A/k} \longrightarrow 0$$

and the left map turns out to be easily computable via the Jacobian matrix. In fact, the left map turns out to be an injection, giving us $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$. Thus, the module $\Omega_{A/k}$ is a "global" way of encoding all the different Zariski cotangent spaces. This is why it is in some sense appropriate to think of $\Omega_{A/k}$ as a "cotangent bundle" whose "fiber" at each closed point $[\mathfrak{m}]$ is $\mathfrak{m}/\mathfrak{m}^2$. We will adopt this point of view in the next section for geometric intuition (and intuition alone; to do this rigorously needs sheaf theory which we shall not develop).

(6) **Universal property of $\Omega_{S/R}$** Let R be a ring, S an R-algebra, an M an S-module.

(a) Show that $\mathrm{Der}_R(S,M)$ and $\Omega_{S/R}$ are S-modules and that $d : S \to \Omega_{S/R}$ is a derivation

(b) Prove the universal property of $\Omega_{S/R}$

(7) **Differentials and localization.** We will show that Ω commutes with localization, in an intuitive way via the "quotient rule." Let S be an R-algebra and U a multiplicatively closed subset of S.

(a) For $u \in U$, use the product rule to determine what $d(1/u)$ should be without a fraction in the derivation

(b) Construct a map $\Omega_{U^{-1}S/R} \to U^{-1}S \otimes_S \Omega_{S/R} \cong U^{-1}\Omega_{S/R}$

(c) Construct the inverse map $U^{-1}\Omega_{S/R} \to \Omega_{U^{-1}S/R}$

(d) Conclude that $\Omega_{U^{-1}S/R} \cong U^{-1}S \otimes_S \Omega_{S/R}$

(8) **The Conormal sequence.** Consider the following situation: we have a ring map $R \to S$ and another map $S \to T$. This exhibits S and T as R-algebras and T as an S-algebra, giving us three modules of differentials: $\Omega_{S/R}$, $\Omega_{T/R}$ and $\Omega_{T/S}$. The latter two are T-modules, while the first is an S-module. To get a T-module out of it, we can consider $T \otimes_S \Omega_{S/R}$.

(a) Show that there is an exact sequence of T-modules

$$T \otimes_S \Omega_{S/R} \xrightarrow{c \otimes db \mapsto cdb} \Omega_{T/R} \xrightarrow{dc \mapsto dc} \Omega_{T/S} \longrightarrow 0$$

This is known as the **relative cotangent sequence** or the **first fundamental exact sequence**. (Hint: $\Omega_{T/S}$ and $\Omega_{T/R}$ are generated by the same elements, but there are extra relations in $\Omega_{T/S}$ since the differential kills everything in S (not just everything in R). For any $b \in S$, $db = 0$, but these are exactly the generators of $\Omega_{S/R}$ as an S-module, hence $T \otimes_S \Omega_{S/R}$ as a T-module)

(b) Suppose the second map in the given situation was a surjective map $\pi : S \to T$ of R-algebras. Show that $\Omega_{T/S} = 0$

(c) Continuing from the previous part, show that there is a well-defined map $I/I^2 \to T \otimes_S \Omega_{S/R}$ given by $f \mapsto 1 \otimes df$

(d) Show that the cokernel of this map is precisely $\Omega_{T/R}$, with the map as given in the first fundamental exact sequence. Conclude that there is an exact sequence

$$I/I^2 \xrightarrow{f \mapsto 1 \otimes df} T \otimes_S \Omega_{S/R} \xrightarrow{c \otimes db \mapsto cdb} \Omega_{T/R} \longrightarrow 0$$

This is known as the **conormal sequence**, or the **second fundamental exact sequence**

(9) **Some useful computations.**

(a) Show that $\Omega_{k[x_1,\ldots,x_n]/k}$ is the free $k[x_1,\ldots,x_n]$-module with generators dx_i

(b) Let K be a separable algebraic extension of k. Show that $\Omega_{K/k} = 0$. This is the main reason we will often assume k is perfect or has characteristic zero.

(c) Show that $\Omega_{k(t_1,\ldots,t_d)/k}$ is a $k(t_1,\ldots,t_d)$-vector space of dimension d

(d) Let K be a separable transcendental extension of k. Assume the fact that there exists a **separating transcendence basis**, i.e. a set of algebraically independent $t_1,\ldots,t_d \in K$, such that $d = \mathrm{trdeg}_k(K)$ and K is algebraic over $K' = k(t_1,\ldots,t_n)$. Show that $\Omega_{K/k}$ is a K-vector space of dimension d.

(10) **Cotangents.** Let A be an integral k-algebra and let $\mathfrak{m}$ be a k-rational maximal ideal. Assume that k is perfect.

(a) What do you get if you apply the conormal sequence to the surjection $\pi : A \to A/\mathfrak{m}$ of k-algebras?

(b) Show that the first map in this sequence is an injection. (Hint: this is equivalent to showing that $df \in \mathfrak{m}\Omega_{A/k} \iff f \in \mathfrak{m}^2$. Now recall the proof showing the Zariski cotangent space is dual to derivations.)

(c) Conclude that $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$

(11) **The Jacobian matrix.** Let A be an integral k-algebra, and let k be perfect. It turns out that if we write $S = k[x_1,\ldots,x_n]$, and $A = S/I$ where $I = \langle f_1,\ldots,f_r\rangle$, the first map in the conormal sequence $I/I^2 \to A \otimes_{k[x_1,\ldots,x_n]} \Omega_{k[x_1,\ldots,x_n]/k} \to \Omega_{A/k} \to 0$ has a nice explicit description via the Jacobian matrix $J = [\partial f_i/\partial x_j]$, giving us an explicit description of $\Omega_{A/k}$. Tensoring with $k = A/\mathfrak{m}$ then lets us compute the dimension of $\mathfrak{m}/\mathfrak{m}^2$.

(a) Consider the conormal sequence associated to the surjection $S \to A$. What is $A \otimes \Omega_{S/k}$? Using this description, describe the map $I/I^2 \to A \otimes \Omega_{S/k}$

(b) Compute $\Omega_{A/k}$ for $A = k[x_1,\ldots,x_n]/\langle f_1,\ldots,f_r\rangle$

(c) Let $\mathfrak{m}$ be a maximal ideal and give $\kappa(\mathfrak{m})$ the structure of an A-module via the map $A \to A/\mathfrak{m}$. Tensor the conormal sequence we have with $\kappa(\mathfrak{m})$ to get a new sequence. Explain why it is exact, and what the terms and maps are.

(d) Prove the **Jacobian criterion for smoothness**: At a point $p = [\mathfrak{m}]$, the dimension of the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is the co-rank of the Jacobian matrix $J(p) = [\partial f_i/\partial x_j]|_p$.

We will see in the next section why the above result is called the Jacobian criterion for *smoothness*.

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4. Smoothness

We have now defined tangents and cotangents for an algebraic variety, and for a given closed point $p = [\mathfrak{m}]$, we have found multiple ways of computing the dimension of the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$. This

number is called the *embedding dimension* at p, and has several characterizations:

$$\boxed{\operatorname{embeddim}_k(A,[\mathfrak{m}]) := \dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim_k k \otimes \Omega_{A/k} = \operatorname{corank} J([\mathfrak{m}])}$$

Recalling the example at the start of this worksheet, we calculated the embedding dimension of the cuspidal cubic at various points. At most points it was 1, but at the origin it was 2. Intuitively, the origin is a *singular* point of the variety, because a curve should one dimension worth of tangents but the tangent space at the origin had 'extra' tangents.

To make the above idea rigorous, we should first define the 'expected' dimension of a variety algebraically: this is the Krull dimension of the coordinate ring. Then, we can define a point to be *singular* if the associated Zariski cotangent space has a different embedding dimension that the dimension of a variety; a point that is not singular is called *smooth* or *regular*. While this may seem arbitrary, there are several reasons why this definition is 'right':

- The embedding dimension at a point is always at least the dimension of the coordinate ring, so a point is singular precisely when it has *more* tangents than expected (it cannot have less)
- Every maximal ideal of the coordinate ring has the same height, so smoothness can be checked locally: a point is smooth if and only if the local ring at that point satisfies $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \operatorname{ht}(\mathfrak{m})$. Recalling that this is the condition for $A_{\mathfrak{m}}$ to be a regular local ring, we say that a point is smooth if and only if the associated local ring is regular. This explains why smoothness is also called regularity.
- 'Most' points on a variety are smooth. The points where the embedding dimension agrees with the dimension of a variety is a Zariski open set (which is always dense). This is the algebraic analogue of Sard's theorem from differential geometry, and it means that singular points are always the exception.

We will prove the above claims below. Before we do this, we will fix some terminology and notation to appeal to some geometric intuition. We will work over A a finitely-generated integral k-algebra over a perfect field k, which in particular covers the algebraically closed case. We will write $\Omega := \Omega_{A/k}$, and write $\Omega/\mathfrak{m}$ for the module $\Omega/\mathfrak{m}\Omega \cong k \otimes \Omega$; we will call this the *fiber* of Ω over $\mathfrak{m}$. The localization $\Omega_{\mathfrak{m}}$ will be called the *stalk* at $\mathfrak{m}$. Using Nakayama's lemma, our argument for generic smoothness will be the algebraic analogue of the geometric fact that if a set is a basis for the fiber of a vector bundle over a point, it can be extended to a set of spanning local sections in a neighborhood of that point, giving a bound on the fiber dimension in that neighborhood. Of course, this is just the analogy, and we will only use module theory for the proof.

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(12) **The dimension of a variety**. Let k be a perfect field and A a finitely-generated integral k-algebra of dimension d. Our goal here is to show that for any maximal ideal$\mathfrak{m}$ of A, $\operatorname{ht}(\mathfrak{m}) = d$, and that $\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) \geq d$, justifying a 'local' check for smoothness.

(a) Let $B = k[y_1, \ldots, y_d]$ be a Noether normalization of A. Let $\mathfrak{m}$ is a maximal ideal of A. Justify the statement: $\mathfrak{m} \cap B$ is a maximal ideal of B with the same height as that of $\mathfrak{m}$

(b) Show that every maximal ideal in B has height d.

(c) Conclude that every maximal ideal of A has height equal to $\dim(A)$.

(d) Use Krull's height theorem to show that $\dim_{\kappa(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 \geq \text{height}(\mathfrak{m}) = \dim(A)$.

(e) Show that the variety associated to A is smooth at point $[\mathfrak{m}]$ if and only if $A_{\mathfrak{m}}$ is a regular local ring. Thus, even though smoothness was originally defined using a global quantity ($\dim A$), it can be checked locally

(13) **Geometric Nakayama**. Let M be a finitely generated module over a ring R. Let $\mathfrak{p}$ be a prime of R such that the fiber of M at $\mathfrak{p}$, defined as $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}}$, has dimension n as an $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ vector space

(a) Show that there exist n elements $f_1, \ldots, f_n$ that generate $M_{\mathfrak{p}}$ as an $R_{\mathfrak{p}}$ module

(b) Show that there exists a g such that the images of the numerators of $f_1, \ldots, f_n$ generate $M[1/g]$ as a $R[1/g]$-module.

(c) Show that for any $[\mathfrak{q}] \in D(g)$, these elements generate $M_{\mathfrak{q}}$ as a $R_{\mathfrak{q}}$ module

(d) Show that this means $\dim(M_{\mathfrak{q}}/\mathfrak{q}M_{\mathfrak{q}})$ is less than or equal to n, for every $\mathfrak{q} \in D(g)$

(14) **Generic Smoothness.** Armed with the geometric Nakayama, we only need to do the calculation of the fiber dimension of Ω at one convenient point. Let A be a finitely-generated integral k-algebra and $\Omega = \Omega_{A/k}$ the module of differentials. Suppose that k has characteristic zero.

(a) Show that the fiber of Ω at $\langle 0 \rangle$ is $\Omega_{K/k}$

(b) Show that $\Omega_{K/k}$ is a $\dim(A)$-dimensional vector space over K

(c) Use geometric Nakayama to conclude that there is a Zariski open set U such that for any maximal idea $\mathfrak{m} \in U$, $\dim(\mathfrak{m}/\mathfrak{m}^2)$ is exactly $\dim(A)$

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5. References

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